

WESTERN GRAY WOLF CASE FILE

Shared evidence packet - all stakeholder groups receive the same core evidence

In 2020, the U.S. Fish and Wildlife Service reconsidered federal protections for gray wolves in the lower 48 states, excluding the separately listed Mexican wolf. Review the information below before receiving your stakeholder assignment.

Your task: Use the shared evidence below, plus your stakeholder-specific case file, to support or oppose delisting. Distinguish evidence relevant to ESA status from other social, economic, or political concerns.

Population recovery

Gray wolves in the lower 48 states occurred primarily in two large metapopulations: the western United States and the western Great Lakes.

In the West:

- More than 1,900 wolves were documented in 2015.
- Later estimates included approximately 1,000 wolves in Idaho and 819 in Montana.
- Minimum counts documented at least 311 wolves in Wyoming.
- Oregon, Washington, and California collectively contained at least 310 wolves, including wolves inside and outside the previously delisted Northern Rocky Mountains population.
- Idaho and Montana had changed their estimation methods, so their newer estimates were not directly comparable with earlier minimum counts.
- Western U.S. wolves were connected through dispersal to an estimated 15,000 wolves in western Canada. [1, 2]

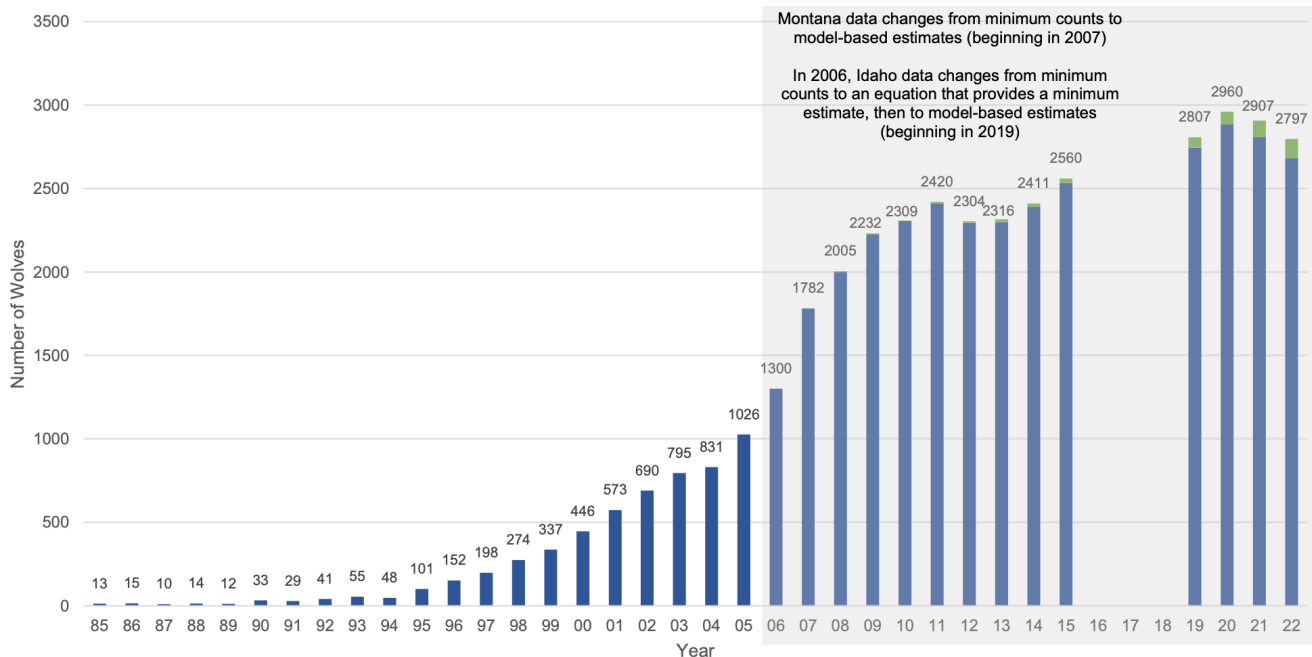


Figure 1: Sourced from Western Gray Wolf SSA, Figure 9. Minimum number of gray wolves counted or estimated in the Western United States, 1985–2022, both inside of the Northern Rocky Mountain (NRM) population (blue) and outside of the NRM (green). Total number of wolves in the Western United States metapopulation indicated at the top of each year's bar. These estimates do not include Mexican wolves. Note that this graph does not include estimates for the total number of wolves in 2016, 2017, or 2018, as we do not have population estimates from Idaho for these years so could not produce a total estimate for the Western United States

Recovery benchmarks

The Northern Rocky Mountains recovery goal called for:

- At least 30 breeding pairs and 300 wolves;
- Equitable distribution among Montana, Idaho, and Wyoming;
- Maintenance of these conditions for three consecutive years; and
- Genetic exchange among the three subpopulations, occurring naturally or through management.

Each state was expected to maintain a buffer of at least 15 breeding pairs and 150 wolves.

The population met its numerical and distributional goals in 2000 and sustained them for three consecutive years by 2002. In 2009, FWS reported at least 45 reproductively successful packs, at least 450 wolves during winter, more than 65,000 square miles of suitable habitat, and genetic exchange among the three recovery areas. [1]

Historical and current range

Gray wolves historically occupied much of the lower 48 states. By the 1970s they remain in only a small fraction of that range. Their occupied range expanded substantially after 1978 but remained smaller and less widely distributed than their historical range. [1, 2]

FWS stated that losing historical range can reduce population size, geographic redundancy, ecological representation, and connectivity. The agency also stated that historical-range loss must be considered alongside the species' current condition, the causes of the loss, and whether those causes continue to affect the population. [1]

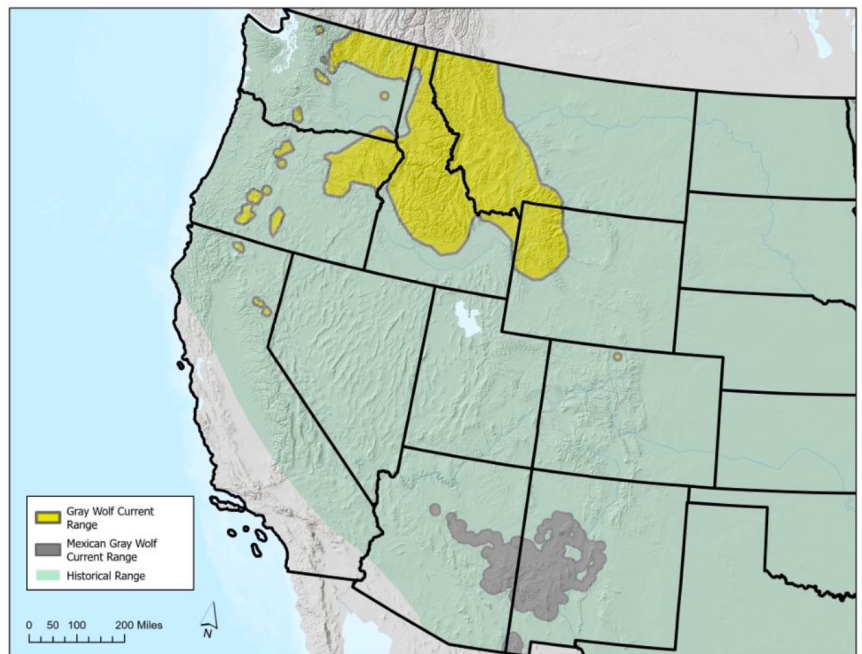


Figure 2: Sources from the Western Gray Wolf SSA, Figure 2. Historical (green) and current (yellow) gray wolf range in the Western United States. The U.S. portion of Mexican wolf range is depicted in gray.

Human-caused mortality

Human-caused mortality was identified as the principal cause of the historical decline of gray wolves. Government-funded eradication programs, poisoning, trapping, shooting, and bounties eliminated wolves from much of their former range.

After federal and state protections restricted intentional killing, wolf abundance and occupied range increased. Outside the Northern Rocky Mountains reintroduction areas, FWS identified regulation of human-caused mortality as the primary reason for this increase.

Studies reviewed by FWS produced several findings:

- Human-caused mortality can add directly to natural mortality.
- Its effect can change with wolf density and local conditions.
- Pack dissolution and breeder loss do not always produce lasting population-level declines.
- Reproduction, dispersal, and reduced natural mortality can compensate for some additional deaths.
- Increasing mortality can reduce wolf abundance, dispersal, recolonization, and connectivity.

- FWS anticipated that lethal control and public harvest would increase after delisting. The agency concluded that moderate increases were unlikely to cause dramatic declines across the populations it evaluated.

Genetic diversity and connectivity

A Minnesota study compared wolves before and after one year of public harvest. It reported:

- No detectable change in genetic diversity;
- A slight increase in broad-scale genetic structure; and
- Changes in the geographic pattern of successful dispersal.

The study covered only two years, limiting conclusions about long-term changes or their causes.

FWS anticipated that delisting could alter genetic diversity or population structure, particularly in smaller peripheral populations. The agency also concluded that continued dispersal—even at a lower rate—would likely maintain genetic diversity in the larger metapopulations. [1]

Management after federal protection

Gray wolves in the Northern Rocky Mountains had already experienced periods of state management before the 2020 rule.

FWS reported that:

- The Northern Rocky Mountains population remained above federal recovery levels during post-delisting monitoring.
- Montana, Idaho, and Wyoming each maintained state wolf-management programs.
- State, Tribal, and federal agencies managed wolf populations above established population targets.
- State-authorized hunting could occur on many state and federal public lands.
- National parks and national wildlife refuges continued to provide protected habitat, although wolves could be exposed to hunting or other mortality after leaving those lands.
- A population decline, failure to maintain state-level buffers, or a significant change in state law or management objectives could trigger another federal status review.

Sources

[1] U.S. Fish and Wildlife Service, 2020 final rule, Federal Register.

[2] U.S. Fish and Wildlife Service. 2023. Species Status Assessment for the Gray Wolf (*Canis lupus*) in the Western United States, Version 1.2. Especially Executive Summar and Chapters 2-4.